

BA 951 PhD Finance I Problem Sets

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1. No Arbitrage

Assume there are two possible states of the world: s_1 and s_2 . There are two assets, a risk-free asset returning R_f in each state, and a risky asset with initial price equal to 1 and date-1 payoff x . Let $R_d = x(s_1)$ and $R_u = x(s_2)$. Assume without loss of generality that $R_u > R_d$.

- (a) What conditions on R_f , R_d and R_u are equivalent to the absence of arbitrage opportunities?
- (b) Assuming the conditions from the previous part hold, compute the unique vector of state prices, and compute the unique risk-neutral probabilities of states s_1 and s_2 .
- (c) Suppose another asset is introduced into the market that pays $\max(x - K, 0)$ for some constant K . Compute the price at which this asset should trade, assuming the conditions from part (a) hold.

2. State Prices and Stochastic Discount Factor

Assume there are three possible states of the world: s_1 , s_2 , and s_3 . Assume there are two assets: a risk-free asset returning R_f in each state, and a risky asset with return R_1 in state s_1 , R_2 in state s_2 , and R_3 in state s_3 . Assume the probabilities are $1/4$ for state s_1 , $1/2$ for state s_2 , and $1/4$ for state s_3 . Assume $R_f = 1.0$, and $R_1 = 1.1$, $R_2 = 1.0$, and $R_3 = 0.9$.

- (a) Prove that there are no arbitrage opportunities.
- (b) Describe the family of state-price vectors $\mathbf{q} = (q_1, q_2, q_3)$.
- (c) Describe the family of SDFs

$$\mathbf{m} = (m_1, m_2, m_3)$$

where m_i denotes the value of the SDF in state s_i . Verify that $m_1 = 4$, $m_2 = -2$, $m_3 = 4$ is an SDF.

3. Complete Contingent Claims Equilibrium

Consider a two-period complete contingent claims economy with a continuum of consumers. Each consumer has a random endowment in the second period and consumes **only in the second period**. In the first period, securities are traded but no money changes hands until the second period. All consumers have log utility over second-period consumption.

- (a) Suppose that there are two equally likely states and all consumers' endowments are the same. They are m with probability $1/2$ and $(1 - a)m$ with probability $1/2$, where $0 < a < 1$. Suppose that a claim to the second-period aggregate endowment is traded and that it costs p in either state, payable in the second period. Compute the equilibrium price p and the expected return on the claim.
- (b) Now suppose that there are three states in the second period, with probability $1/2$ all consumers receive m ; with probability $1/2$, a fraction $(1 - b)$ of consumers receive m and a fraction b receive $(1 - a/b)m$. Note that we must have $b > a$ so the individual endowment is always positive and log utility is defined. In the first period, all consumers face the same probability of being in the latter group, but no insurance markets exist through which they can hedge this risk. Compute the expected return on the claim defined above. Is it higher or lower than before? Interpret this result.

(Hint: Compute the expected utility cost and gain of investing in the asset. The equilibrium asset price should be determined by equating the expected utility cost and gain.)

4. Stochastic Discount Factor and Beta Representation

- (a) Suppose you have a mean-variance efficient return R^{mv} and the risk free rate. using the fact that R^{mv} is perfectly correlated with the stochastic discount factor, construct an SDF, m , in terms of R^f and R^{mv} , with no free parameters.
- (b) Using this result, show that any expected return can be described in a single-beta representation using any mean-variance efficient return (except the risk-free rate).

$$\mathbb{E}[R^i] = R^f + \beta_{i,mv} (\mathbb{E}[R^{mv}] - R^f)$$

5. Mehra-Prescott Model

Work through the the Mehra-Prescott Model with i.i.d consumption Growth. Assume investors have CRRA utilities: $u(c) = \frac{c^{1-\gamma}-1}{1-\gamma}$, where γ is the risk-aversion coefficient. The log consumption growth rate follows:

$$\Delta c_t = \mu + \varepsilon_t, \quad \varepsilon_t \sim i.i.d. \mathcal{N}(0, \sigma^2)$$

- (a) Solve for the risk-free rate in this setting.
- (b) Solve for the price-to-consumption ratio (PC ratio). What is the restriction we must impose on the model parameters in order to have a positive PC ratio? (Hint: conjecture and verify that the PC ratio is constant).
- (c) In this model, the aggregate market return equals to the return on the consumption claim. Derive the equity risk premium. (Hint: The equity risk premium is defined as the difference between the expected log return on any asset and the log risk-free interest rate, plus one-half the own variance of the log asset return. The last term comes from Jensen's inequality.)
- (d) Derive the maximum Sharpe ratio of risky assets.

6. Consumption-Based Asset Pricing

Consider an economy with a representative agent who has CRRA utility with coefficient of relative risk aversion γ . The agent receives a nonstorable endowment. The process for the log endowment, or equivalently the log of consumption c , is:

$$\Delta c_{t+1} = \mu + \phi \Delta c_t + u_{t+1}, \quad u_{t+1} \sim i.i.d. \mathcal{N}(0, \sigma_u^2)$$

where the coefficient ϕ may be either positive or negative. If ϕ is positive, then endowment fluctuations are highly persistent; if it is negative, then they have an important transitory component. Assume that consumption and asset returns are jointly lognormal, with constant variances and covariances.

- (a) Use the representative agent's Euler equations to show that the expected log return on any asset is a linear function of the expected growth rate of the endowment. What is the slope coefficient in this relationship?
- (b) Use the representative agent's Euler equations to show that the equity risk premium is proportional to the conditional covariance of the log asset return with consumption growth. What is the slope coefficient in this relationship? (Hint: The equity risk premium is defined as the difference between the expected log return on any asset and the log risk-free interest rate, plus one-half the own variance of the log asset return. The last term comes from Jensen's inequality.)

According to the Campbell-Shiller approximation, the unexpected return on an asset can be

written as

$$r_{t+1} - \mathbb{E}_t[r_{t+1}] = (\mathbb{E}_{t+1} - \mathbb{E}_t) \left[\sum_{j=0}^{\infty} \rho^j \Delta d_{t+1+j} - \sum_{j=1}^{\infty} \rho^j r_{t+1+j} \right]$$

where d_t is the dividend paid on the asset at time t , ρ is a constant.

- (c) Use this expression to calculate the unexpected return on the consumption claim (an asset which pays aggregate consumption as its dividend).
- (d) Use this expression to calculate the unexpected return on a real consol bond which has a fixed real dividend (coupon) each period.
- (e) Calculate the equity premium and the consol bond premium.
- (f) Show that the bond premium has the opposite sign to ϕ and is proportional to the square of γ . Give an economic interpretation of this result.
- (g) Show that the equity premium is always larger than the bond premium, and the difference between them is proportional to γ . Give an economic interpretation of this result.
- (h) Relate your discussion to the empirical literature on the “equity premium puzzle.”

7. Campbell-Shiller Approximation

Derive the Campbell-Shiller approximation for log return. Use it to derive the return innovation decomposition.

8. Present Value

Consider a stock whose log dividend d , follows a random walk with drift:

$$d_{t+1} = \mu + d_t + \varepsilon_{t+1}, \quad \varepsilon_{t+1} \sim \mathcal{N}(0, \sigma^2)$$

Assume that the required log rate of return on the stock is a constant r .

- (a) We use the notation F_t (for “fundamental value”) to denote the expected present value of dividends, discounted using the required rate of return. Show that F_t is a constant multiple of the dividend D_t . Write the ratio F_t/D_t as a function of the parameters of the model.
- (b) Show that another formula for the stock price which gives the same expected rate of return is

$$P_t = F_t + cD_t^\lambda$$

where $\lambda > 0$ is a function of the other parameters of the model. Solve for λ .

9. Present Value and Return Predictability

Suppose that the log “fundamental value” of a stock, v_t , follows the process:

$$v_t = \mu - \left(\frac{1 - \kappa_1}{\kappa_1} \right) (d_t - v_{t-1}) + v_{t-1} + \varepsilon_t$$

where d_t is the log dividend, ε_t is a white noise error term, μ is a constant, κ_1 is the parameter in the Campbell-Shiller log-linearization:

$$r_{t+1} \approx \kappa_0 + \kappa_1 (p_{t+1} - d_{t+1}) + d_{t+1} - p_t$$

- (a) Show that if the price of the stock equals its fundamental value, then the approximate log stock return defined by the CS log-linearization is unpredictable.
- (b) Now suppose that the stock pays dividends according to the rule:

$$d_t = c + \lambda v_{t-1} + (1 - \lambda)d_{t-1} + \eta_t$$

where c and λ are constants ($0 < \lambda < 1$), and η_t is a white noise error uncorrelated with ε_t . Dividends are partially adjusted towards the fundamental value, where the speed of adjustment is given by λ ¹. Show that if the price of the stock equals its fundamental value, then the log price-dividend ratio follows an AR(1) process. What is the persistence of this process as a function of λ and κ_1 ?

- (c) Suppose that the stock price does not equal the fundamental value, but rather satisfies

$$p_t = v_t - \gamma(d_t - v_t)$$

where $\gamma > 0$. That is, price exceeds fundamental value whenever the latter is high relative to dividends. The dividend policy is the same as part (b). Show that the log price-dividend ratio follows an AR(1) process, and the approximate log stock return satisfies the following simple model:

¹Marsh and Merton (1986) have argued for the empirical relevance of this dividend policy.

$$\mathbb{E}_t[r_{t+1}] = r + x_t$$

$$x_{t+1} = \phi x_t + \xi_{t+1}, \quad -1 < \phi < 1$$

where the expected log stock return is a constant r plus an observable zero-mean variable x_t , which follows an AR(1) process. Also show that the optimal predictor of the log stock return, x_t , is a multiple of the log price-dividend ratio.

(d) Derive the innovations in stock returns and innovations in x_t .

10. Optimal Portfolio Choice

Consider the portfolio choice problem in a one-period model. Suppose there is a risk-free asset with return R_f and n risky assets with returns $\mathbf{R}$. The risky returns are joint normally distributed with mean vector $\boldsymbol{\mu}$ and nonsingular covariance matrix $\boldsymbol{\Omega}$. Let ω_f denote the investment in the risk-free asset, let ω_i denote the investment in risky asset i , and let $\boldsymbol{\omega}$ denote the n -dimensional column vector with ω_i as its i th element. Consider an investor who has time-additive utility and CARA utility:

$$u(c) = -e^{-\gamma c}$$

(a) Compute the investor's optimal portfolio of risky assets, $\boldsymbol{\omega}$.

Now, use asset payoffs and prices instead of returns and solving for the optimal number of shares to hold of each asset instead of the optimal amount to invest. Suppose there is a risk-free asset with return R_f and n risky assets with payoffs x_i and prices p_i . Assume the vector $\mathbf{x} = [x_1, x_2, \dots, x_n]'$ is normally distributed with mean $\boldsymbol{\mu}_x$ and nonsingular covariance matrix $\boldsymbol{\Omega}_x$. Let $\mathbf{p} = [p_1, p_2, \dots, p_n]'$. Let θ_i denote the number of shares the investor considers holding of asset i and set $\boldsymbol{\theta} = [\theta_1, \theta_2, \dots, \theta_n]'$.

(b) Compute the investor's optimal vector of share holdings, $\boldsymbol{\theta}$.

11. Mean-Variance Frontier

In this exercise we work through the mathematics of mean-variance efficient sets. Let there be N risky assets with mean vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Omega}$. Denote $\boldsymbol{\iota}$ as a length- N vector of ones. Define $\boldsymbol{\omega}_a$ the $(N \times 1)$ vector of portfolio weights for an arbitrary portfolio a with weights summing to unity. Derive the following things in theory.

(a) The minimum-variance frontier of all risky assets.

(b) Mean, variance, and portfolio weights of the global minimum-variance portfolio.

- (c) The mean-variance frontier (with a risk-free rate).

12. Zero-Beta Portfolio

This exercise introduces the concept of zero-beta portfolios. A portfolio z is a zero-beta portfolio with respect to portfolio p if z has zero covariance with p .

Prove that for any minimum-variance portfolio p (except for the global minimum-variance portfolio), there exists infinite zero-beta portfolios. However, all of these zero-beta portfolios have the same expected return. This return is called the zero-beta rate with respect to portfolio p .

(Hint: The covariance between any two portfolios a and b is $\sigma_{a,b} = \omega'_a \Omega \omega_b$.)

13. CAPM and Stochastic Discount Factor

This exercise relates the stochastic discount factor and beta pricing relationship to the factor models. Let us start with the single-factor model—CAPM. CAPM can be expressed as:

$$E[R_i] = R_f + \beta_i \lambda$$

where $E[\cdot]$ is the expectation operator, R_i is the realized return on asset i , R_f is the risk-free return, β_i is asset i 's market beta, and λ is a positive market risk premium. Now, consider a stochastic discount factor of the form:

$$m = a + bR_m$$

where a and b are constants and R_m is the realized return on the market portfolio. Also, denote the variance of the return on the market portfolio as σ_m^2 .

- (a) Derive an expression for λ as a function of a , b , $\mathbb{E}[R_m]$, and σ_m^2 . (Hint: You may want to start from the equilibrium expression $0 = \mathbb{E}[m(R_i - R_f)]$.)
- (b) Note that the equation $1 = E[mR_i]$ holds for all assets. Consider the case of the risk-free asset and the case of the market portfolio, and solve for a and b as a function of R_f , $\mathbb{E}[R_m]$, and σ_m^2 .
- (c) Using the formula for a and b in part (b), show that $\lambda = \mathbb{E}[R_m] - R_f$.

14. Multi-Factor Models

This exercise works on the general formula for multi-factor models in the vector form. The fundamental equation for risk premium— $E[R_i] - R_f = -R_f \text{Cov}(m, R_i)$ —tells us that the key object in studying risk premium is the covariance term $\text{Cov}(m, R_i)$. However, this is generally impossible to compute because of the extremely high dimension of the calculation. A factor model is basically trying to reduce the dimension and make the problem tractable. A general factor model has the following beta pricing formula in the vector form:

$$\mathbb{E}[\mathbf{R}] - R_f \mathbf{1} = \underbrace{-R_f \text{Cov}(\mathbf{R}, m) \equiv \boldsymbol{\beta} \cdot \boldsymbol{\lambda} \equiv \text{Cov}(\mathbf{R}, \mathbf{F}) \boldsymbol{\Sigma}_F^{-1} \cdot \boldsymbol{\lambda}}_{\text{dimension reduction}} \quad (1)$$

where $\mathbf{R} \in \mathcal{R}^n$ is a vector of returns for N risky assets, $\mathbf{1}$ is an $N \times 1$ vector of ones, $\boldsymbol{\beta}$ is an $N \times K$ matrix capturing the exposure of N assets to a set of K risk factors, $\boldsymbol{\lambda}$ is a $K \times 1$ matrix capturing the prices of K risk factors or factor risk premiums, $\mathbf{F} = \{f_1, f_2, \dots, f_K\}$ are the factors and $\boldsymbol{\Sigma}_F$ denotes the covariance matrix for the factors. By the projection theory, $\boldsymbol{\beta} = \text{Cov}(\mathbf{R}, \mathbf{F}) \boldsymbol{\Sigma}_F^{-1}$.

- (a) Prove that if there is a multi-factor model of equation (1), then there exists an affine SDF of the form:

$$m = a + \mathbf{b}'\mathbf{F} \quad (2)$$

- (b) [Optional] Prove that if there is an SDF of equation (2), then there exists a multi-factor model of equation (1). Part (a) and (b) complete the proof that there is an equivalence between factor models and SDFs.
- (c) [Optional] If an individual factor f_j is a return, then its price of risk λ_j is the risk premium of the return $\mathbb{E}[f_j] - R_f$ (e.g., the market factor in CAPM). If an individual factor is an excess return, then its price of risk is its mean return $\mathbb{E}[f_j]$ (e.g., the value factor in FF3). Prove these two arguments.

15. Term Structure and Expectation Hypothesis

Suppose that the monetary authority controls short-term interest rates by setting:

$$y_t^{(1)} = y_{t-1}^{(1)} + \lambda(y_t^{(2)} - y_t^{(1)}) + \varepsilon_t$$

with $\lambda > 0$. Intuitively, the monetary authority tries to smooth interest rates but raises them when the yield curve is steep. Suppose also that the two-period bond yield satisfies:

$$y_t^{(2)} = \frac{1}{2} \left(y_t^{(1)} + \mathbb{E}_t[y_{t+1}^{(1)}] \right) + x_t$$

where x_t is a term premium that represents the deviation of the two-period yield from the expectations hypothesis. The variable x_t follows an AR(1) process:

$$x_t = \phi x_{t-1} + \eta_t$$

The error terms ε_t , and η_t are serially uncorrelated and uncorrelated with each other.

- (a) Assume $\lambda\phi < 2$, show that this model can be solved for an interest-rate process of the form:

$$y_t^{(1)} = y_{t-1}^{(1)} + \gamma x_t + \varepsilon_t$$

Express the coefficient γ as a function of the other parameters in the model.

- (b) The expectations hypothesis of the term structure is often tested by regressing the scaled change in the short rate onto the yield spread

$$\left(y_{t+1}^{(1)} - y_t^{(1)} \right) / 2 = \alpha + \beta \left(y_t^{(2)} - y_t^{(1)} \right) + u_{t+1}$$

and testing the hypothesis that the coefficient $\beta = 1$. If the model described above holds, what is the population value of the regression coefficient β ?

- (c) [Optional] Now consider a version of the problem involving n -period bonds. The monetary authority sets short-term interest rates as:

$$y_t^{(1)} = y_{t-1}^{(1)} + \lambda(y_t^{(n)} - y_t^{(1)}) + \varepsilon_t$$

The n -period bond yield is determined by:

$$y_t^{(n)} - y_t^{(1)} = (n-1) \left(\mathbb{E}_t[y_{t+1}^{(n)}] - y_t^{(n)} \right) + x_t$$

where x_t now measures the deviation of the n -period yield from the expectations hypothesis. (This formulation ignores the distinction between $y_t^{(n-1)}$ and $y_t^{(n)}$.) As before, x_t follows an AR(1) process. What is the coefficient γ in this case? What is the regression coefficient β in the following regression that is also used to test the expectation hypothesis?

$$(n-1) \left(y_{t+1}^{(n)} - y_t^{(n)} \right) = \alpha + \beta \left(y_t^{(n)} - y_t^{(1)} \right) + u_{t+1}$$

- (d) [Optional] Do you find the model you have studied in this problem to be a plausible explanation of empirical findings on testing the expectation hypothesis?

16. A Single-Factor Affine Term Structure Model

In this exercise, we work through the simplest affine term structure model (ATSM) with a single state variable, which follows a homoscedastic AR(1) process (Vasicek, 1977):

$$m_{t+1} = -x_t - \frac{1}{2} \left(\frac{\lambda}{\sigma} \right)^2 - \left(\frac{\lambda}{\sigma} \right) \varepsilon_{t+1}$$

$$x_{t+1} = \mu + \phi x_t + \sigma \varepsilon_{t+1}$$

In this model, the log SDF m_{t+1} is linear in the AR(1) state variable x_t , ε_{t+1} follows a standard normal distribution. Notice that m_{t+1} and x_{t+1} are exposed to the same source of risk ε_{t+1} . Parameter σ governs volatility and λ governs the price of risk.

- (a) Show that the state variable x_t in this model is exactly the short rate (one-period risk-free rate).
- (b) Solve the bond prices and bond yields as functions of the state variable.
- (d) Compute the bond risk premium. Are (real) bonds hedges or risky assets in this model?

17. Long Run Risks

In this exercise, we work through the “Case I” long run risks model in Bansal and Yaron (2004), where there are fluctuations only in the expected growth rates (constant volatility).

Investors have Epstein-Zin preferences:

$$U_t = \left\{ (1 - \delta) C_t^{\frac{1-\gamma}{\theta}} + \delta \left(\mathbb{E} [U_{t+1}^{1-\gamma}] \right)^{\frac{1}{\theta}} \right\}^{\frac{\theta}{1-\gamma}}$$

where δ is the rate of time preference, $\theta = \frac{1-\gamma}{1-1/\psi}$ where $\gamma > 1$ is the risk aversion coefficient, $\psi > 1$ is the intertemporal elasticity of substitution (IES). Suppose $r_{a,t+1}$ is the log return on a consumption claim that delivers aggregate consumption as its dividends each period, and g_{t+1} is the log growth rate of aggregate consumption. Log dividend growth rate is modeled separately and denoted as $g_{d,t+1}$. $r_{m,t+1}$ denotes the log market return on the dividend claim.

In this setup, we can derive the log SDF as:

$$m_{t+1} = \theta \log \delta - \frac{\theta}{\psi} g_{t+1} + (\theta - 1) r_{a,t+1}$$

The model can be characterized by the following equations:

$$g_{t+1} = \mu + x_t + \sigma \eta_{t+1} \tag{1}$$

$$g_{d,t+1} = \mu_d + \phi x_t + \varphi_d \sigma u_{t+1} \tag{2}$$

$$x_{t+1} = \rho x_t + \varphi_e \sigma e_{t+1} \tag{3}$$

where the three shocks $\eta_{t+1}, u_{t+1}, e_{t+1} \sim \text{i.i.d. } \mathcal{N}(0, 1)$. Two additional parameters, $\phi > 1$ and $\psi_d > 1$, allow us to calibrate the overall volatility of dividends (which in the data are significantly larger than that of consumption) and its correlation with consumption. A key feature of long run risks models is that the consumption growth and dividend growth contain a small persistent predictable component x_t .

To demonstrate the mechanisms and intuitions of the model, we can derive analytical solutions via the Campbell-Shiller approximation:

$$r_{a,t+1} = \kappa_0 + \kappa_1 z_{t+1} - z_t + g_{t+1}$$

$$r_{m,t+1} = \kappa_{m,0} + \kappa_{m,1} z_{m,t+1} - z_{m,t} + g_{d,t+1}$$

where z_t and $z_{m,t}$ correspond to the log price–consumption ratio and log price–dividend ratio.

- (a) Solve z_t and $z_{m,t}$ as functions of the state variable x_t . Discuss how the persistent expected growth (x_t) affects the two price ratios.
- (b) Derive the innovations to the SDF.
- (c) Compute the market risk premium.

18. Habit Model

Suppose habit accumulation is linear, and there is a constant risk-free rate or linear technology equal to the discount rate, $R^f = 1/\beta$. The consumer's problem is:

$$\begin{aligned}
& \max \sum_{t=0}^{\infty} \beta^t \frac{(C_t - X_t)^{1-\gamma}}{1-\gamma} \\
& s.t. \quad \sum_t C_t = \sum_t e_t + W_0 \\
& s.t. \quad X_t = \theta \sum_{j=0}^{\infty} \phi^j C_{t-j}
\end{aligned}$$

where e_t is a stochastic endowment. In an internal habit specification, the consumer considers all the effects that current consumption has on future utility through X_{t+j} . In an external habit specification, the consumer ignores such terms. Show that the two specifications give identical asset pricing predictions in this simple model, by showing that internal-habit marginal utility is proportional to external-habit marginal utility, state by state.

19. Production-based Model

The linear quadratic permanent income model is a very useful general equilibrium model that we can solve in closed form. It specifies a production technology rather than fixed endowments, and it easily allows aggregation of disparate consumers².

The consumer maximizes:

$$\mathbb{E} \sum_{t=0}^{\infty} \beta^t \left(-\frac{1}{2} \right) (c_t - c^*)^2$$

subject to a linear technology:

$$\begin{aligned}
k_{t+1} &= (1+r)k_t + i_t \\
i_t &= e_t - c_t
\end{aligned}$$

e_t is an exogenous endowment or labor income stream. Assume $\beta = 1/(1+r)$: the discount rate equals the interest rate or marginal productivity of capital.

²Hansen (1987) is a wonderful exposition of what one can do with this setup.

(a) Show that optimal consumption follows:

$$c_t = rk_t + r\beta \sum_{j=0}^{\infty} \beta^j \mathbb{E}_t[e_{t+j}] \quad (4)$$

$$c_t = c_{t-1} + (\mathbb{E}_t - \mathbb{E}_{t-1}) r\beta \sum_{j=0}^{\infty} \beta^j e_{t+j} \quad (5)$$

i.e., consumption equals permanent income, precisely defined, and consumption follows a random walk whose innovations are equal to innovations in permanent income.

(b) Assume that the endowment e_t follows an AR(1) process:

$$e_t = \rho e_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim i.i.d. \mathcal{N}(0, \sigma_\varepsilon^2)$$

Specialize equations (4) and (5). Discuss how the persistence of income, ρ , affects the consumption functions.

(c) Keep assuming that the endowment e_t follows an AR(1) process. Calculate the one period interest rate and the price of a claim to the consumption stream. Interpret the time-variation in the price of the consumption stream. (Hint: The price of a claim to the consumption stream can be written as a function of c_t and σ_ε^2 . This consumer gets more risk averse as consumption rises to c^* . c^* is the bliss point, so at the bliss point there is no average return that can compensate the consumer for greater risk.)